

Relevance-Based Prediction: Factor Selection for Saugus Schools

Four outcomes: MCAS grades 3–8, Dropout Rate, MCAS grade 10 ELA, Education Budget Share

One RBP run per outcome on ALL candidate variables — no feature pruning (faithful to Kritzman)

Exhibit 5 importance is descriptive: positive = helps, ≈ 0 = benign / diversified away (fn. 12), < 0 = harmful

Saugus analyzed as the prediction task; most/least relevant towns identified per Exhibit 4

Leave-one-out LOO r validates the predictions across all MA districts

Model Summary (all candidates used — no pruning)			
MCAS Grades 3-8	Dropout Rate	MCAS Grade 10 (ELA)	Education Budget Share
Predicted on 17 candidates	Predicted on 18 candidates	Predicted on 18 candidates	Predicted on 18 candidates
Positive importance: 17 of 17	Positive importance: 17 of 18	Positive importance: 17 of 18	Positive importance: 18 of 18
LOO r = +0.944	LOO r = +0.831	LOO r = +0.896	LOO r = +0.897

Pre-Specified Candidate Pool — Feature Reduction

Features selected on domain grounds before running RBP. Reduces K/N ratio from ~0.19 to ~0.09 (target ≈ 0.085).

Kept — 14 pure predictors

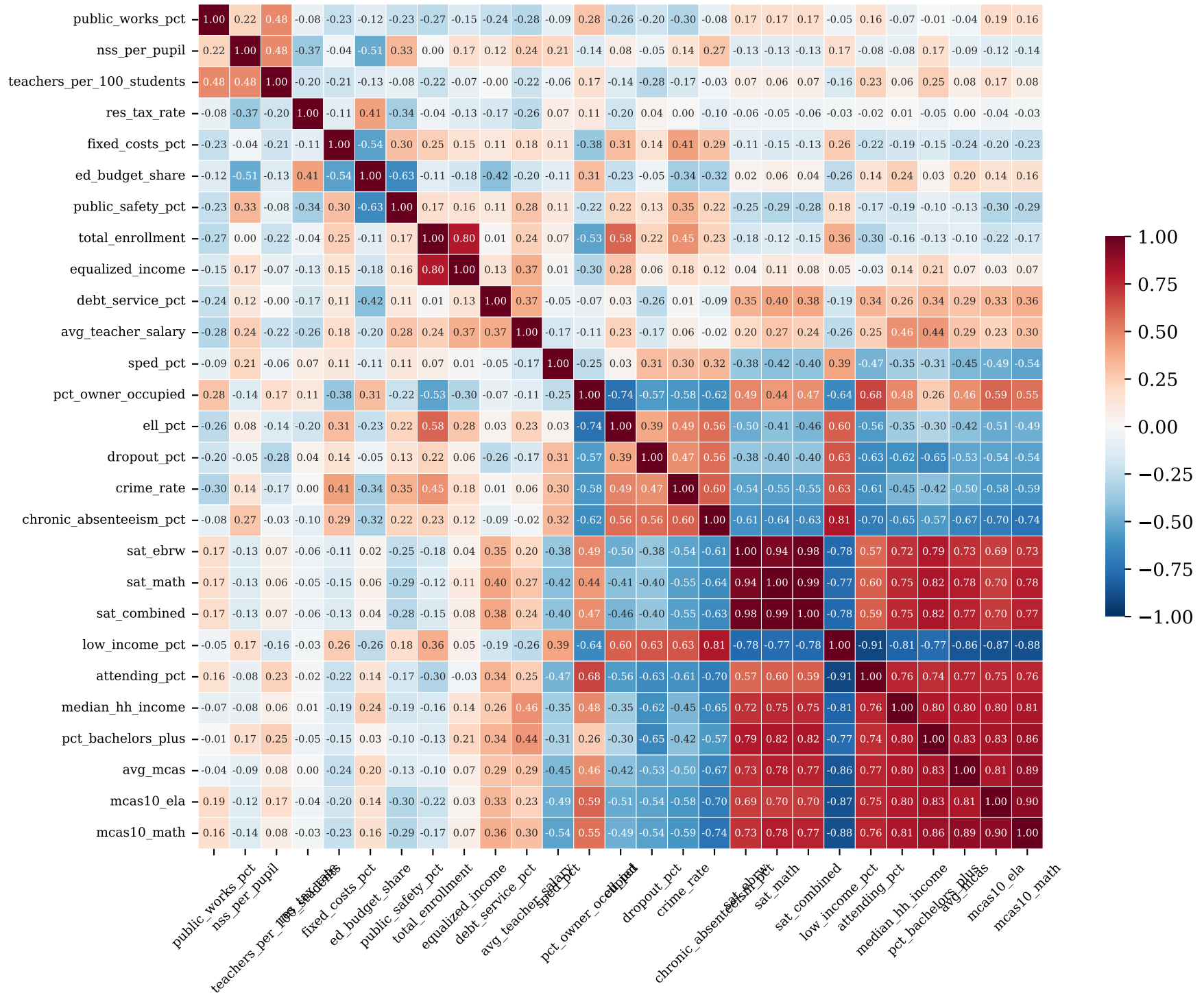
Group	Feature	Measures
Poverty	low_income_pct	% low-income students
Wealth	median_hh_income	Median household income
Wealth	equalized_income	Equalized property value/capita
Human capital	pct_bachelors_plus	% adults with bachelor's+
Housing	pct_owner_occupied	% owner-occupied homes
Community	crime_rate	Crime incidents per capita
Community	res_tax_rate	Residential tax rate
Engagement	chron_absenteeism_pct	% chronically absent
Demographics	ell_pct	% English language learners
Demographics	sped_pct	% special education students
Size	total_enrollment	District enrollment
Staffing	teachers/100_students	Teachers per 100 students
Staffing	avg_teacher_salary	Average teacher salary
Spending	nss_per_pupil	Net school spending/pupil

Dropped — 12 redundant/problematic features

Dropped	Kept instead	Reason
high_needs_pct	low_income_pct	$r=+0.981$ — near-duplicate
acs_poverty_pct	low_income_pct	$r=+0.87$ — redundant poverty
foundation_budget_pp	nss_per_pupil	$r=+0.92$ — redundant spending
ch70_per_pupil	equalized_income	$r\approx-0.87$ — inverse wealth proxy
teacher_fte	total_enrollment	$r=+0.994$ — linear with enroll
total_population	total_enrollment	$r=+0.953$ — size proxy
teachers_per_100_fte	t_per_100_students	$r=+0.872$ — duplicate ratio
teacher_spending_pp	nss_per_pupil	$r\approx+0.85$ — redundant spending
gf_exp_per_capita	res_tax_rate	$r\approx+0.75$ — redundant municipal
pct_65_plus	—	Weak signal, no policy lever
com_tax_rate	res_tax_rate	$r\approx+0.65$ — commercial tax dup
mcas10_math	mcas10_elar	$r=+0.90$ — one grade-10 measure

Correlation threshold for deduplication: $|r| > 0.87$. Cross-outcome features (avg_mcas, mcas10_ela, dropout_pct, attending_pct) handled separately per model via exclusion rules.

Feature Correlation Matrix — All Candidates (hierarchically clustered)
Features ordered by |correlation| — dark red/blue clusters = high redundancy

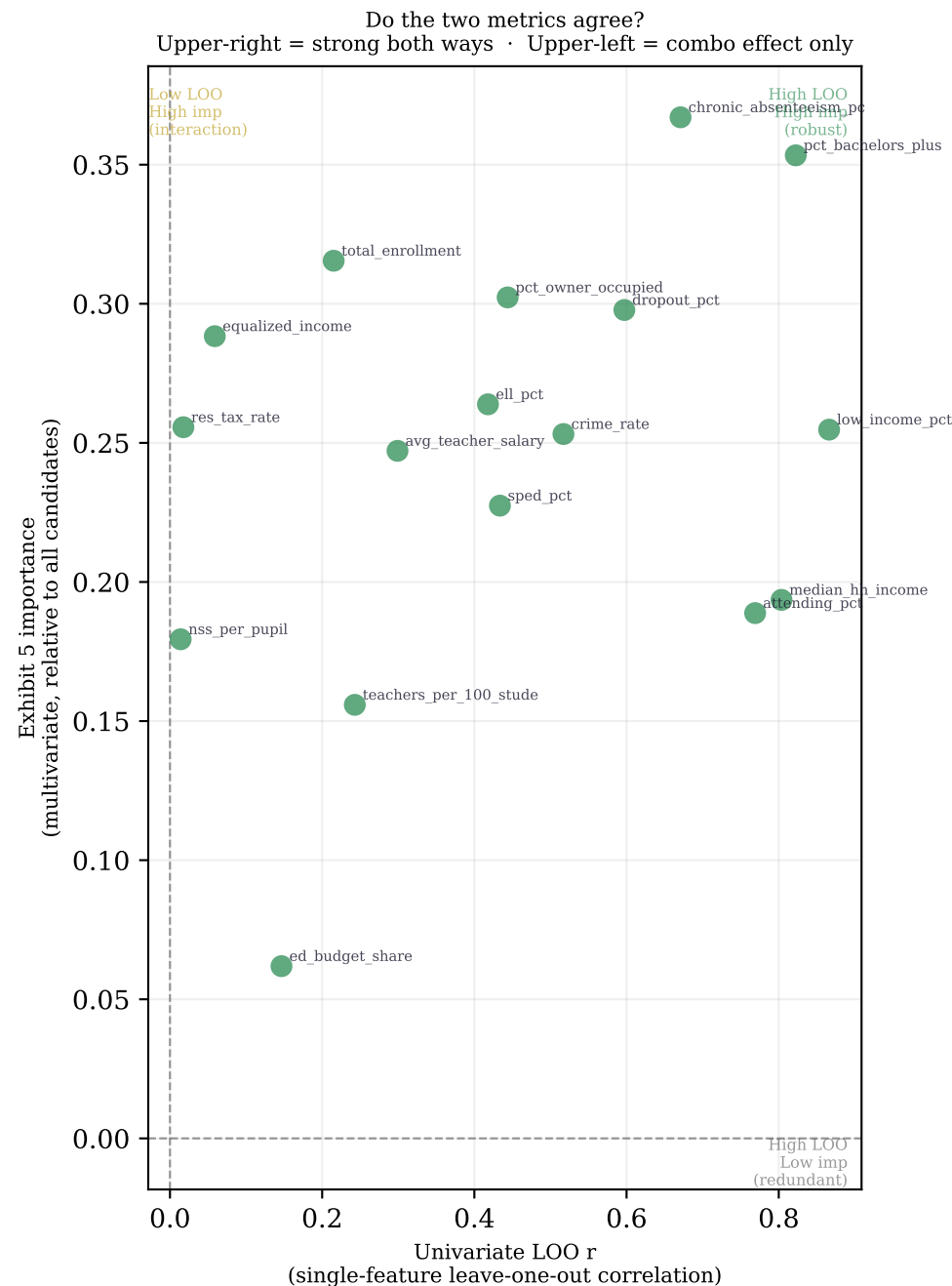
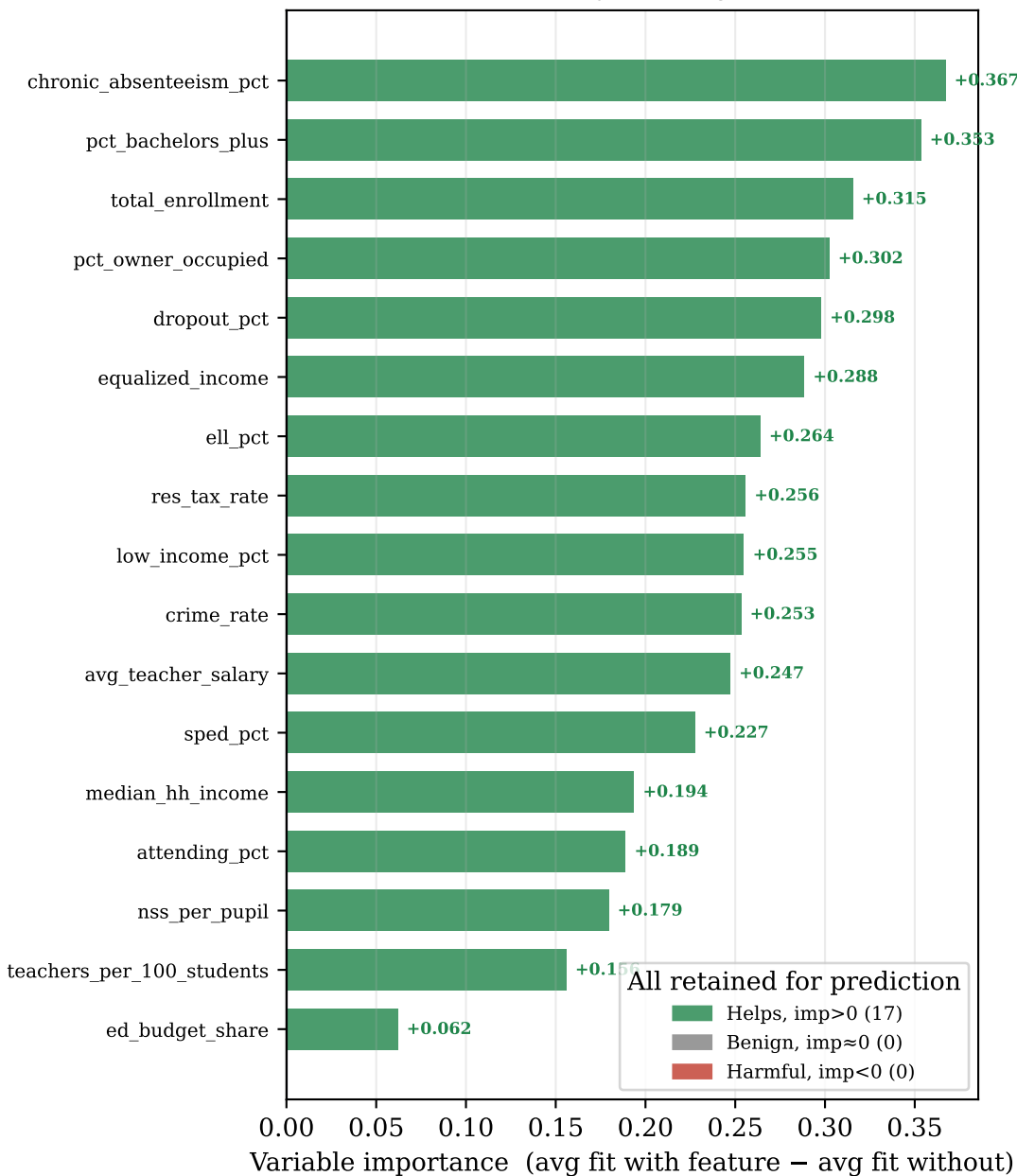


Pearson r. Hierarchical average-linkage clustering on $|1-r|$ distance. Features with $|r| > 0.87$ collapsed to one representative in the candidate pool.

Variable Importance (Exhibit 5): MCAS Grades 3-8

17 candidates, all used for prediction · n_random=3000 · top-3 NOT seed-stable · 17 help (imp>0) · 0 benign (≈0) · 0 harmful (imp<0)

All Candidates — Variable Importance (all retained)
Green = helps (>0) · Grey ≈ 0 benign · Red < 0 harmful

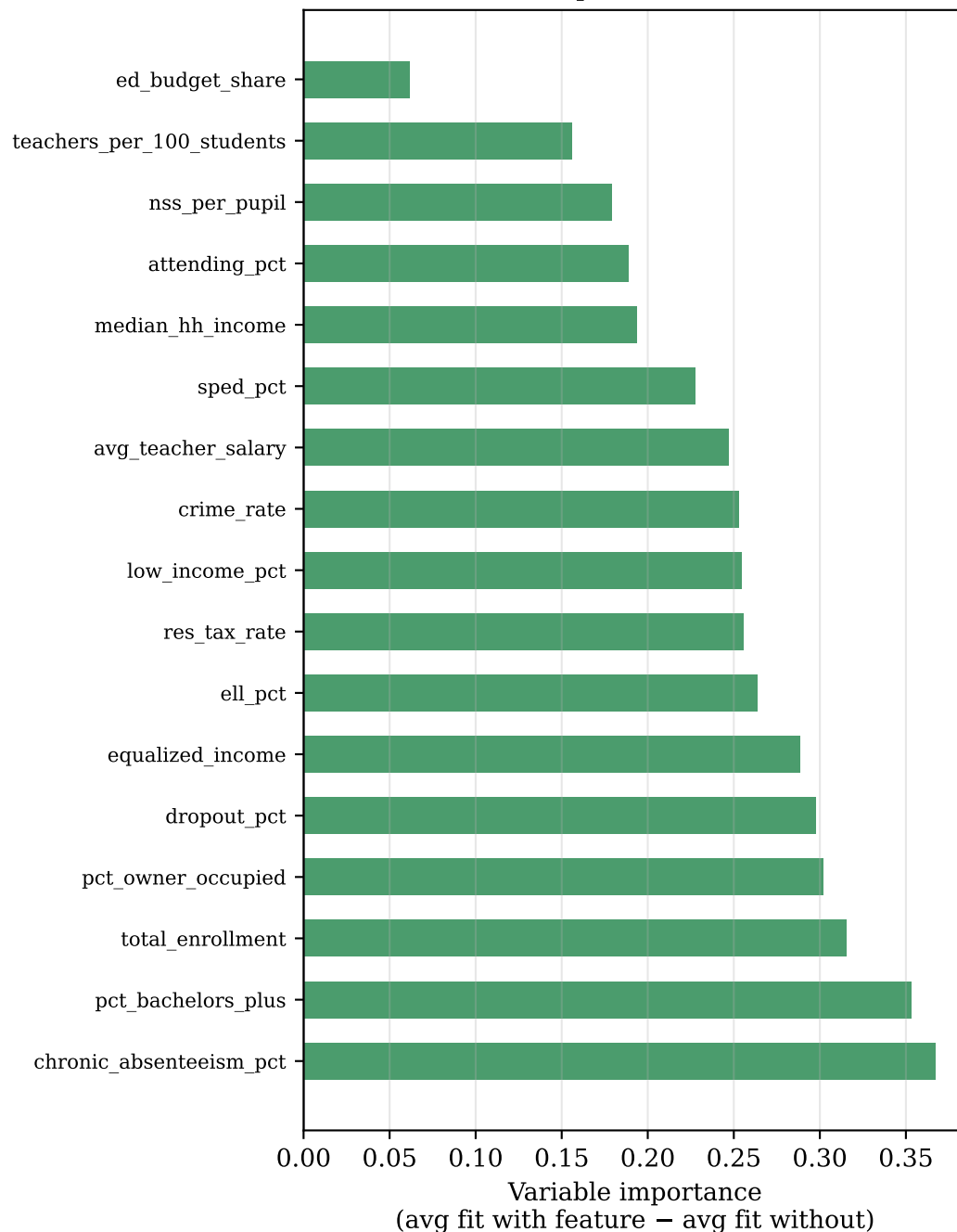


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Saugus RBP Analysis: MCAS Grades 3-8

Prediction task = Saugus · Predicted: 21.8pp Actual: 29.0pp Gap: +7.2pp (Fit: 0.1561)

Variable Importance (Exhibit 5)



Most Relevant Towns (highest weight)

Town	Weight	avg_mcas
Everett	0.0772	0.18
Gardner	0.0674	0.21
Dracut	0.0620	0.38
Pittsfield	0.0614	0.24
Agawam	0.0536	0.34

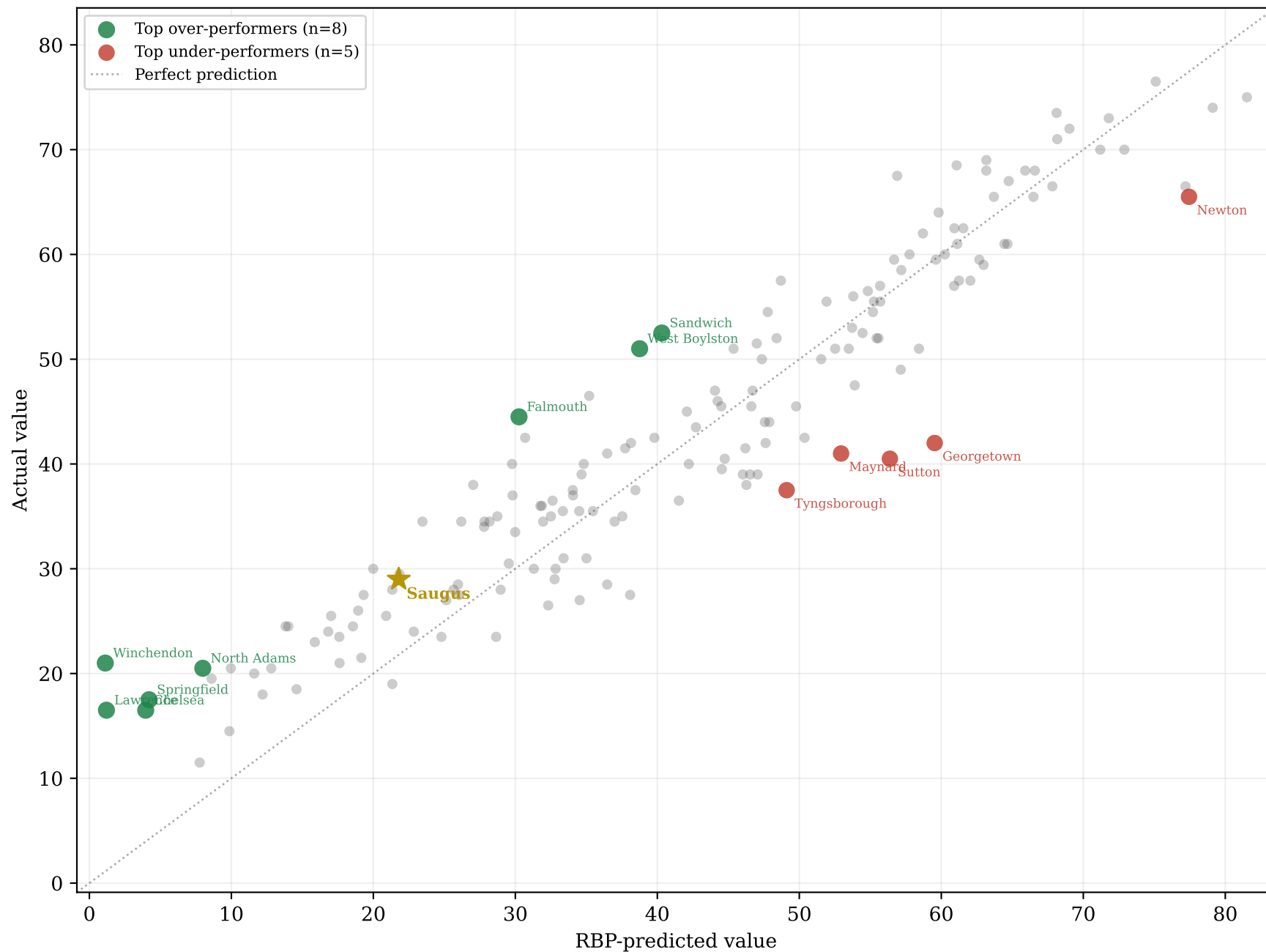
Least Relevant Towns (lowest weight)

Town	Weight	avg_mcas
Foxborough	-0.0174	0.51
Hingham	-0.0173	0.73
Newburyport	-0.0172	0.49
Andover	-0.0169	0.66
Marshfield	-0.0164	0.55

Over- and Under-Performers: MCAS Grades 3-8

Saugus gap: +7.2pp · Gold star = Saugus · Green = over-performers · Red = under-performers

Actual vs Predicted — MCAS Grades 3-8 (LOO r = 0.944)



Residual = actual - predicted. Positive = performing better than demographics suggest.

What Overachievers Do Differently: MCAS Grades 3-8

Feature values for top overachievers vs Saugus — ordered by RBP variable importance

Feature values — Saugus (highlighted) vs overachievers
Ordered by |importance|, top 15 shown

Feature	Saugus	Winchendon	Lawrence	Falmouth	Imp
chronic_absenteeism_pct	31.2	35.6	29.2	24.0	+0.367
pct_bachelors_plus	31.4	23.9	16.4	51.9	+0.353
total_enrollment	3,084	1,360	15,500	3,130	+0.315
pct_owner_occupied	73.9	77.8	30.3	82.7	+0.302
dropout_pct	2.90	7.50	7.80	1.00	+0.298
equalized_income	1.17B	278M	1.77B	1.68B	+0.288
ell_pct	13.0	1.90	42.1	7.50	+0.264
res_tax_rate	10.7	12.5	9.25	6.28	+0.256
low_income_pct	46.6	56.8	85.1	39.5	+0.255
crime_rate	2,813	3,287	3,435	3,500	+0.253
avg_teacher_salary	89,778	83,722	84,516	92,617	+0.247
sped_pct	18.8	27.9	19.4	24.4	+0.227
median_hh_income	100,819	85,160	57,903	92,862	+0.194
attending_pct	64.4	36.4	43.4	68.0	+0.189
nss_per_pupil	17,369	19,559	23,178	25,271	+0.179

Overachiever residuals vs Saugus

District	Actual	Predicted	Gap
>> Saugus	29.0	21.8	+7.2pp
Winchendon	21.0	1.1	+19.9pp
Lawrence	16.5	1.2	+15.3pp
Falmouth	44.5	30.3	+14.2pp
Springfield	17.5	4.2	+13.3pp
Chelsea	16.5	4.0	+12.5pp
North Adams	20.5	8.0	+12.5pp

Gap = actual – predicted. Positive = outperforming demographic expectation. Feature values shown in raw units; importance scores from RBP Exhibit 5.

What This Means: MCAS Grades 3-8

Predicted 21.8pp · Actual 29.0pp · Gap +7.2pp (higher is better) · Saugus ranks 28 of 168 MA districts on this measure

The Bottom Line

Saugus's MCAS grades 3–8 proficiency (ELA + Math, meeting/exceeding) is 29.0%, 7.2pp better than its demographic prediction of 21.8% — ranking 28 of 168 MA districts, in the top 25%. A district with Saugus's income, poverty, and enrollment profile is predicted to land near this level.

What Shapes Saugus's Predicted Peer Group

Top RBP Exhibit 5 drivers of the Saugus prediction, vs. the statewide median:

Factor	Saugus	MA median	Importance
Students chronically absent (10%+)	31.2%	15.8%	+0.37
Adults with a bachelor's degree+	31.4%	46.9%	+0.35
Total district enrollment	3,084	1,669	+0.32
Owner-occupied housing units	73.9%	78.3%	+0.30
Annual dropout rate	2.9%	0.9%	+0.30

Importance measures how much each factor sharpens the search for Saugus's true comparison districts — not whether higher values are good or bad for the outcome itself.

What Over-Performing Towns Do Differently

Same factors, for the 8 MA districts that most exceed their own prediction:

Factor	Saugus	Peer median	Difference
Students chronically absent (10%+)	31.2%	29.8%	-1.4pp
Adults with a bachelor's degree+	31.4%	25.5%	-6.0pp
Total district enrollment	3,084	2,761	-323
Owner-occupied housing units	73.9%	65.5%	-8.4pp
Annual dropout rate	2.9%	3.8%	+0.9pp

Difference = over-performer median – Saugus. Sign direction depends on the factor; see takeaway below.

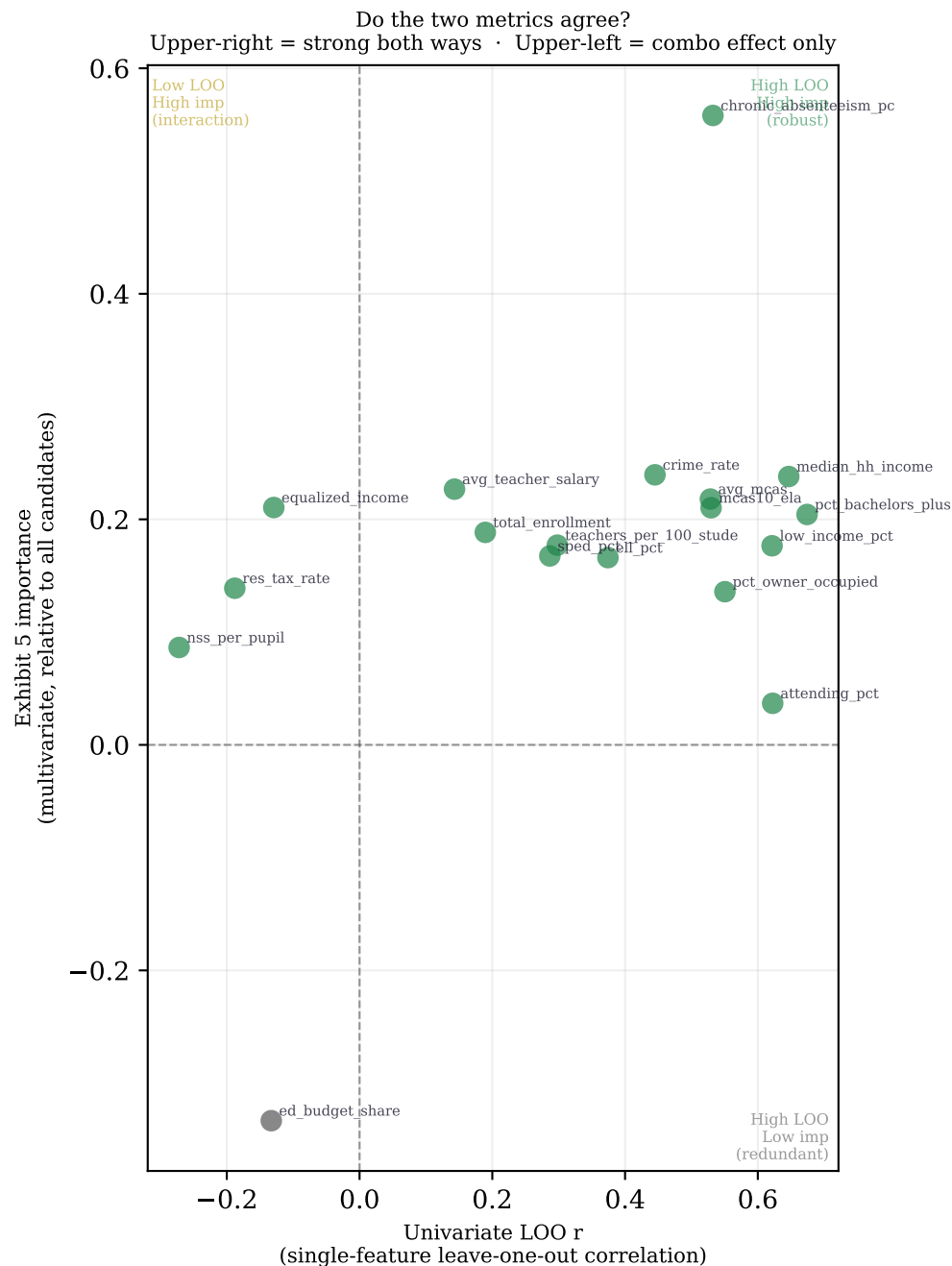
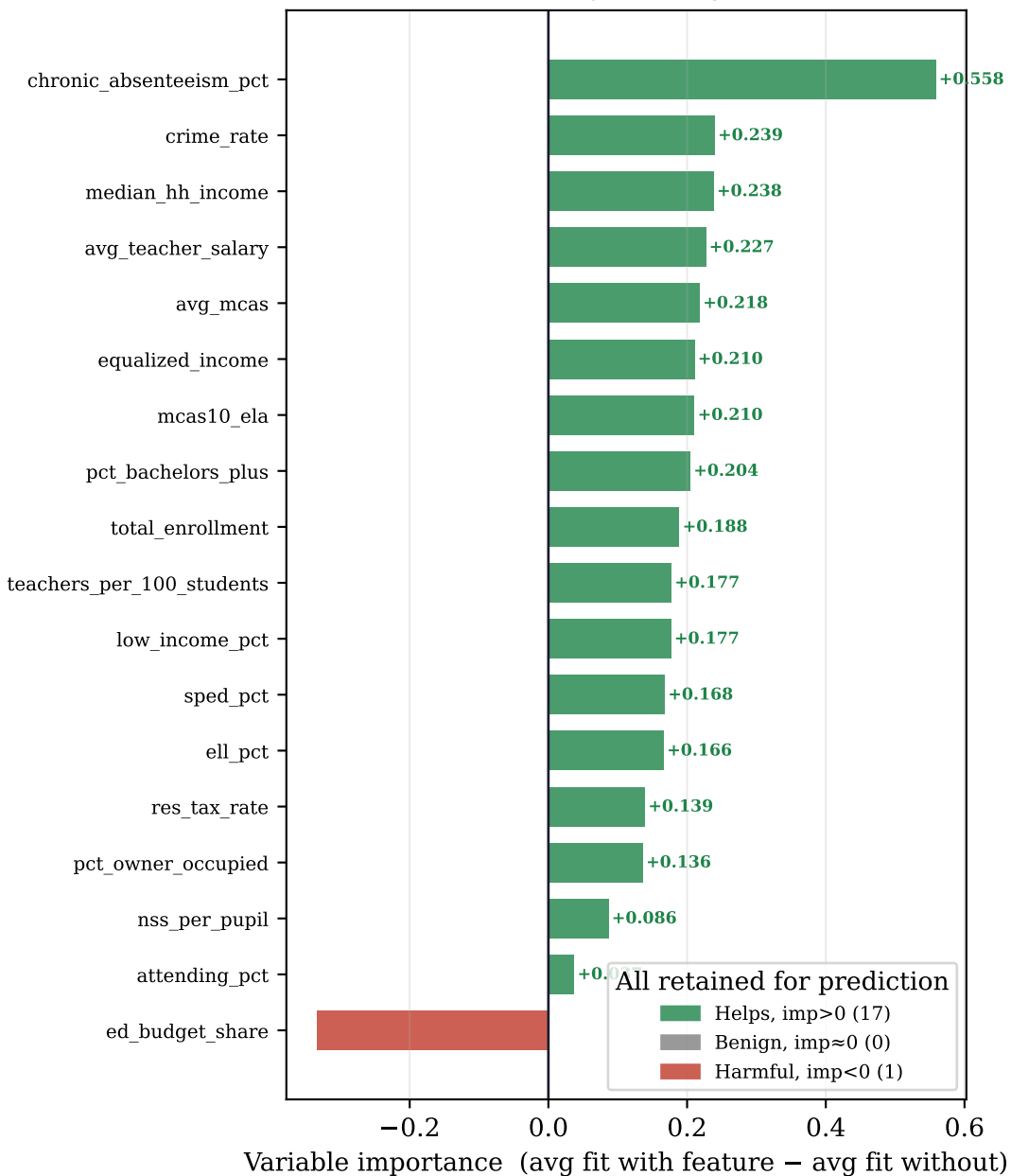
The Takeaway

Saugus does better here than its demographic profile predicts. The strongest RBP driver of Saugus's predicted peer group is students chronically absent (10%+) (31.2% vs the 15.8% state median); the widest gap to the over-performing districts is total district enrollment (-323, peer median – Saugus). (Driver ranking is close enough at the top that the #1 factor is not fully seed-stable — read the top group, not the exact order.) The drivers below are where Saugus differs most from the towns that beat their own predictions — the most plausible places to move this result.

Variable Importance (Exhibit 5): Dropout Rate

18 candidates, all used for prediction · n_random=3000 · top-3 NOT seed-stable · 17 help (imp>0) · 0 benign (≈0) · 1 harmful (imp<0)

All Candidates — Variable Importance (all retained)
Green = helps (>0) · Grey ≈ 0 benign · Red < 0 harmful

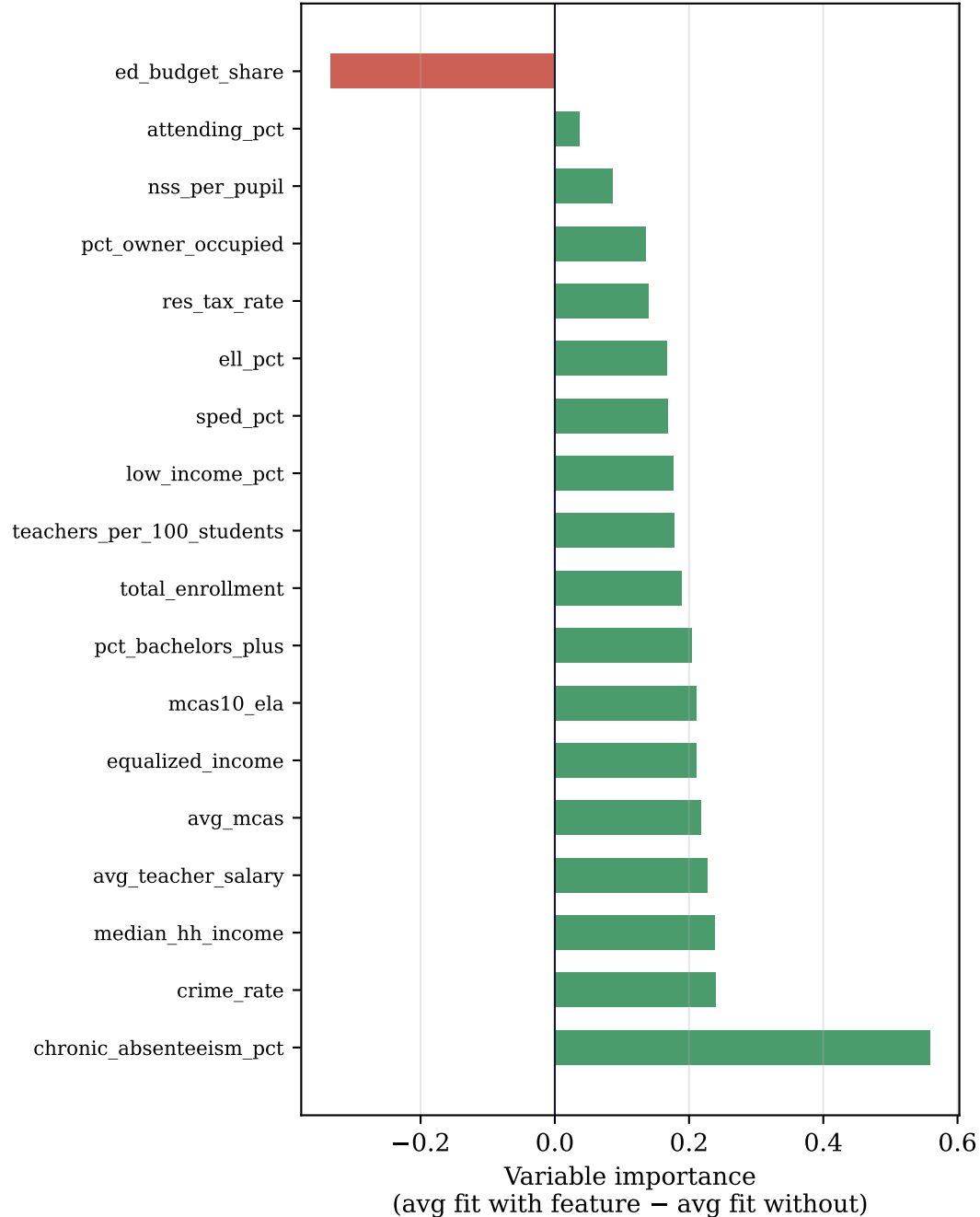


ng feature – avg fit of cells without it. Univariate LOO r: single-feature leave-one-out correlation. One RBP run per outcome on all candidates — importance is descriptive, no feature is dropped (Kritzman fn. 12), so this ch

Saugus RBP Analysis: Dropout Rate

Prediction task = Saugus · Predicted: 3.7pp Actual: 2.9pp Gap: -0.8pp (Fit: 0.1543)

Variable Importance (Exhibit 5)



Most Relevant Towns (highest weight)

Town	Weight	dropout_pct
Everett	0.0774	5.20
Pittsfield	0.0690	4.80
Gardner	0.0667	2.90
Dracut	0.0605	1.40
Agawam	0.0599	1.30

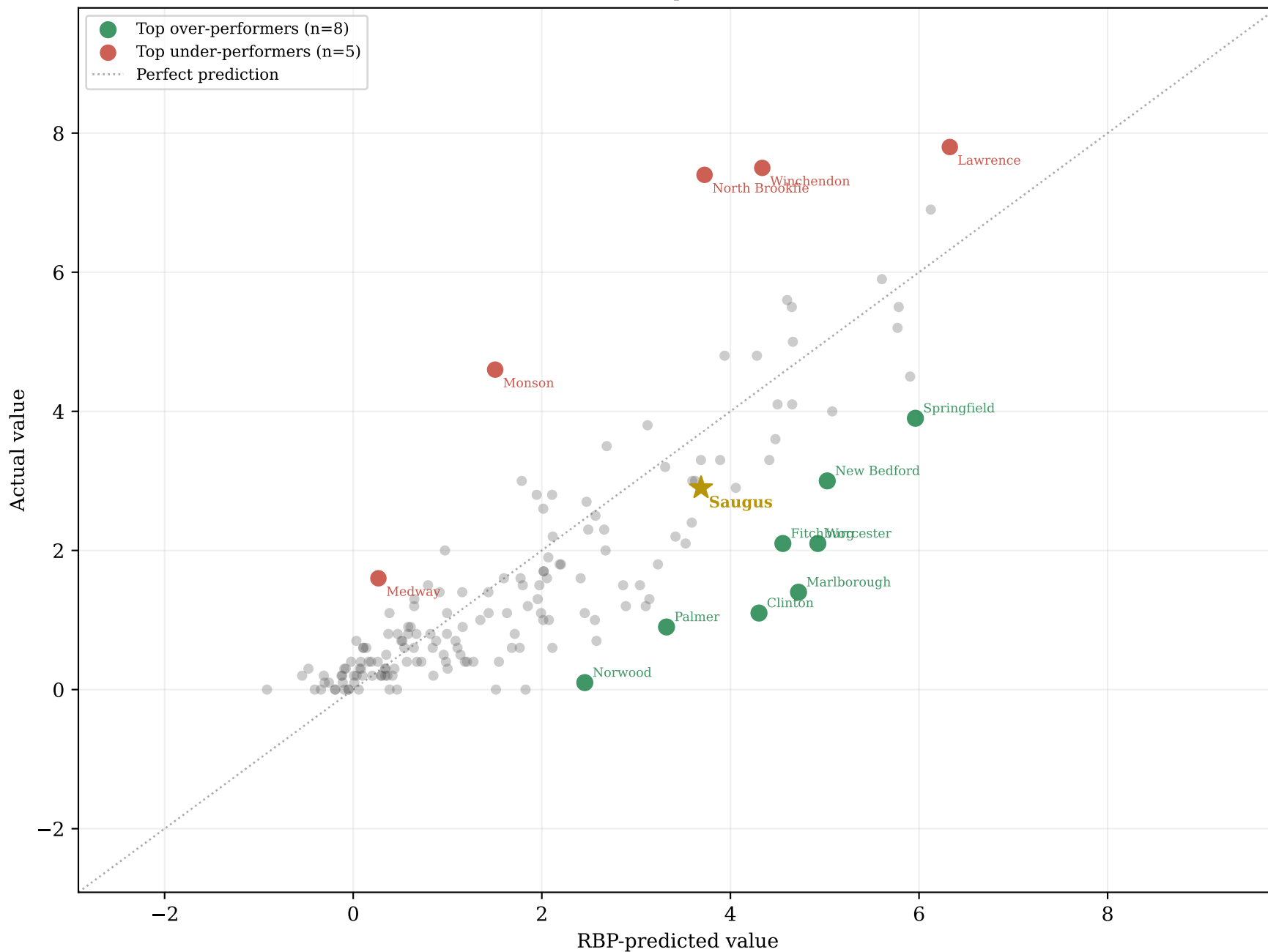
Least Relevant Towns (lowest weight)

Town	Weight	dropout_pct
Andover	-0.0166	0.20
Natick	-0.0160	0.70
Hingham	-0.0160	0.00
Marshfield	-0.0159	0.30
Newburyport	-0.0157	0.00

Over- and Under-Performers: Dropout Rate

Saugus gap: -0.8pp · Gold star = Saugus · Green = over-performers · Red = under-performers

Actual vs Predicted — Dropout Rate (LOO r = 0.831)



What Overachievers Do Differently: Dropout Rate

Feature values for top overachievers vs Saugus — ordered by RBP variable importance

Feature values — Saugus (highlighted) vs overachievers
Ordered by |importance|, top 15 shown

Feature	Saugus	Marlborou	Clinton	Worcester	Imp
chronic_absenteeism_pct	31.2	23.5	7.70	28.3	+0.558
ed_budget_share	32.3	43.0	49.6	45.8	-0.333
crime_rate	2,813	3,695	324.5	5,437	+0.239
median_hh_income	100,819	95,047	93,849	67,544	+0.238
avg_teacher_salary	89,778	87,213	87,059	101,529	+0.227
avg_mcas	0.29	0.24	0.30	0.23	+0.218
equalized_income	1.17B	1.65B	514M	5.00B	+0.210
mcas10_ela	0.47	0.40	0.43	0.40	+0.210
pct_bachelors_plus	31.4	40.6	35.1	34.1	+0.204
total_enrollment	3,084	5,572	2,237	27,606	+0.188
teachers_per_100_student	5.87	7.36	7.31	7.40	+0.177
low_income_pct	46.6	58.3	52.7	71.4	+0.177
sped_pct	18.8	16.4	21.9	21.9	+0.168
ell_pct	13.0	32.9	19.7	30.4	+0.166
res_tax_rate	10.7	10.2	13.1	13.8	+0.139

Overachiever residuals vs Saugus

District	Actual	Predicted	Gap
>> Saugus	2.9	3.7	-0.8pp
Marlborough	1.4	4.7	+3.3pp better
Clinton	1.1	4.3	+3.2pp better
Worcester	2.1	4.9	+2.8pp better
Fitchburg	2.1	4.6	+2.5pp better
Palmer	0.9	3.3	+2.4pp better
Norwood	0.1	2.5	+2.4pp better

Gap = actual – predicted. Positive = outperforming demographic expectation. Feature values shown in raw units; importance scores from RBP Exhibit 5.

What This Means: Dropout Rate

Predicted 3.7pp · Actual 2.9pp · Gap -0.8pp (lower is better) · Saugus ranks 39 of 168 MA districts on this measure

The Bottom Line

Saugus's dropout rate is 2.9%, 0.8pp better than its demographic prediction of 3.7% — ranking 39 of 168 MA districts, in the top 25%. Dropout reflects re-engagement and attendance recovery as much as demographics.

What Shapes Saugus's Predicted Peer Group

Top RBP Exhibit 5 drivers of the Saugus prediction, vs. the statewide median:

Factor	Saugus	MA median	Importance
Students chronically absent (10%+)	31.2%	15.8%	+0.56
Ed budget share	32.3	48.2	-0.33
Crime incidents per 100k residents	2,813	2,241	+0.24
Median household income	\$100,819	\$109,970	+0.24
Average teacher salary	\$89,778	\$90,677	+0.23

Importance measures how much each factor sharpens the search for Saugus's true comparison districts — not whether higher values are good or bad for the outcome itself.

What Over-Performing Towns Do Differently

Same factors, for the 8 MA districts that most exceed their own prediction:

Factor	Saugus	Peer median	Difference
Students chronically absent (10%+)	31.2%	26.7%	-4.5pp
Ed budget share	32.3	48.9	+16.6
Crime incidents per 100k residents	2,813	4,392	+1,579
Median household income	\$100,819	\$74,724	-26,095
Average teacher salary	\$89,778	\$87,136	-2,642

Difference = over-performer median – Saugus. Sign direction depends on the factor; see takeaway below.

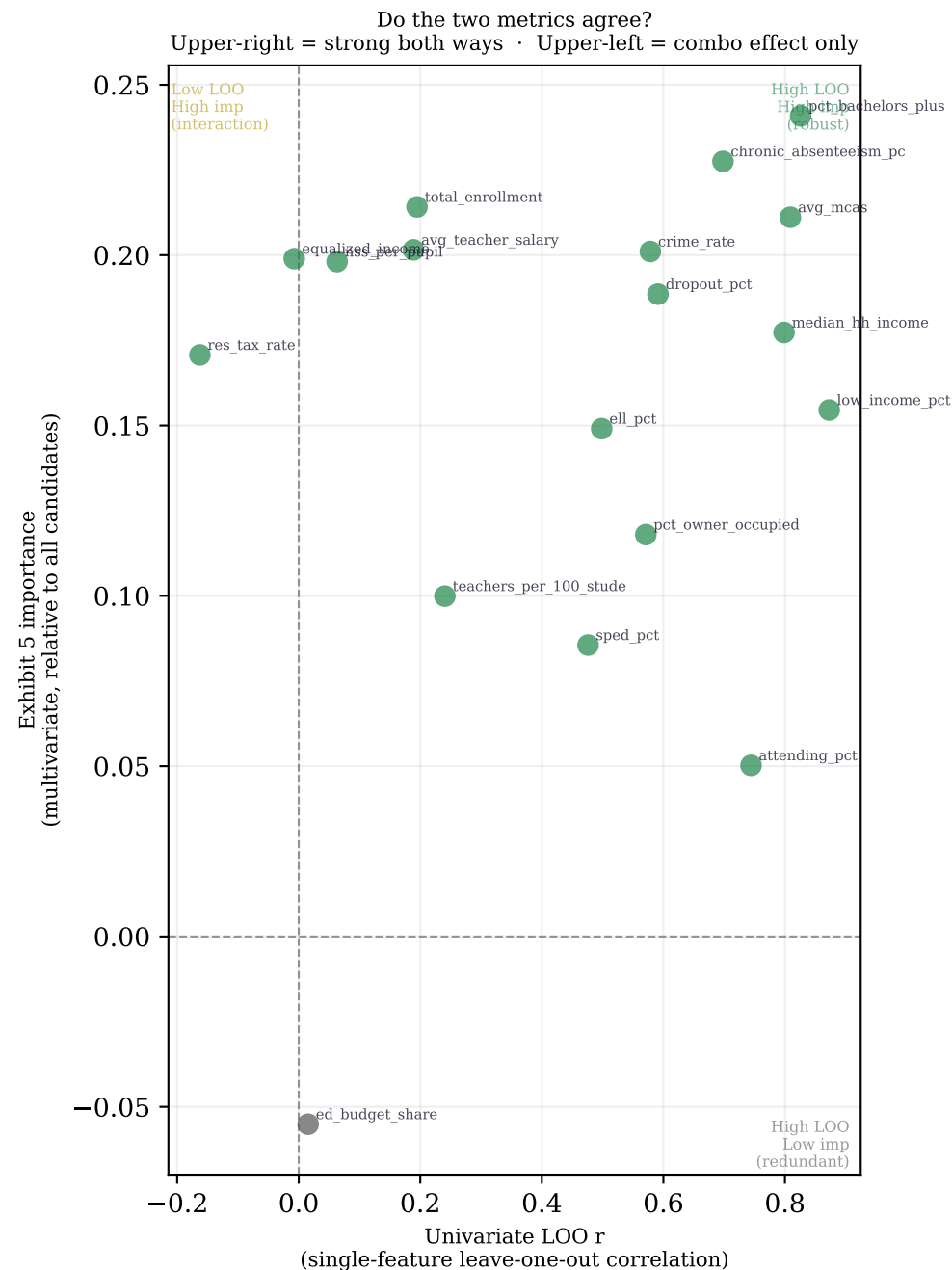
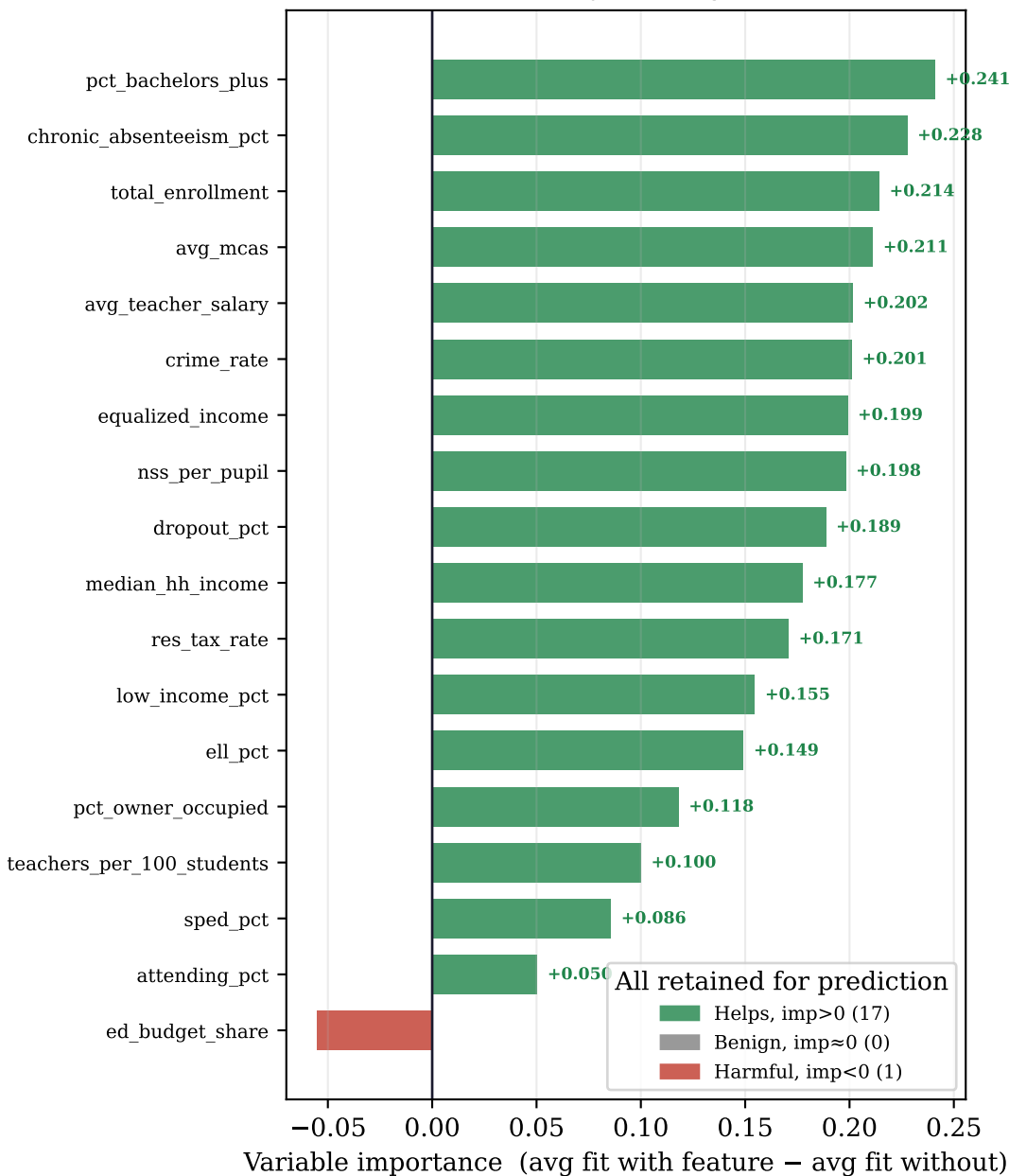
The Takeaway

Saugus does better here than its demographic profile predicts. The strongest RBP driver of Saugus's predicted peer group is students chronically absent (10%+) (31.2% vs the 15.8% state median); the widest gap to the over-performing districts is median household income (-26,095, peer median – Saugus). (Driver ranking is close enough at the top that the #1 factor is not fully seed-stable — read the top group, not the exact order.) Chronic absenteeism is the factor to watch — the main risk to this result if it climbs.

Variable Importance (Exhibit 5): MCAS Grade 10 (ELA)

18 candidates, all used for prediction · n_random=3000 · top-3 seed-stable · 17 help (imp>0) · 0 benign (≈0) · 1 harmful (imp<0)

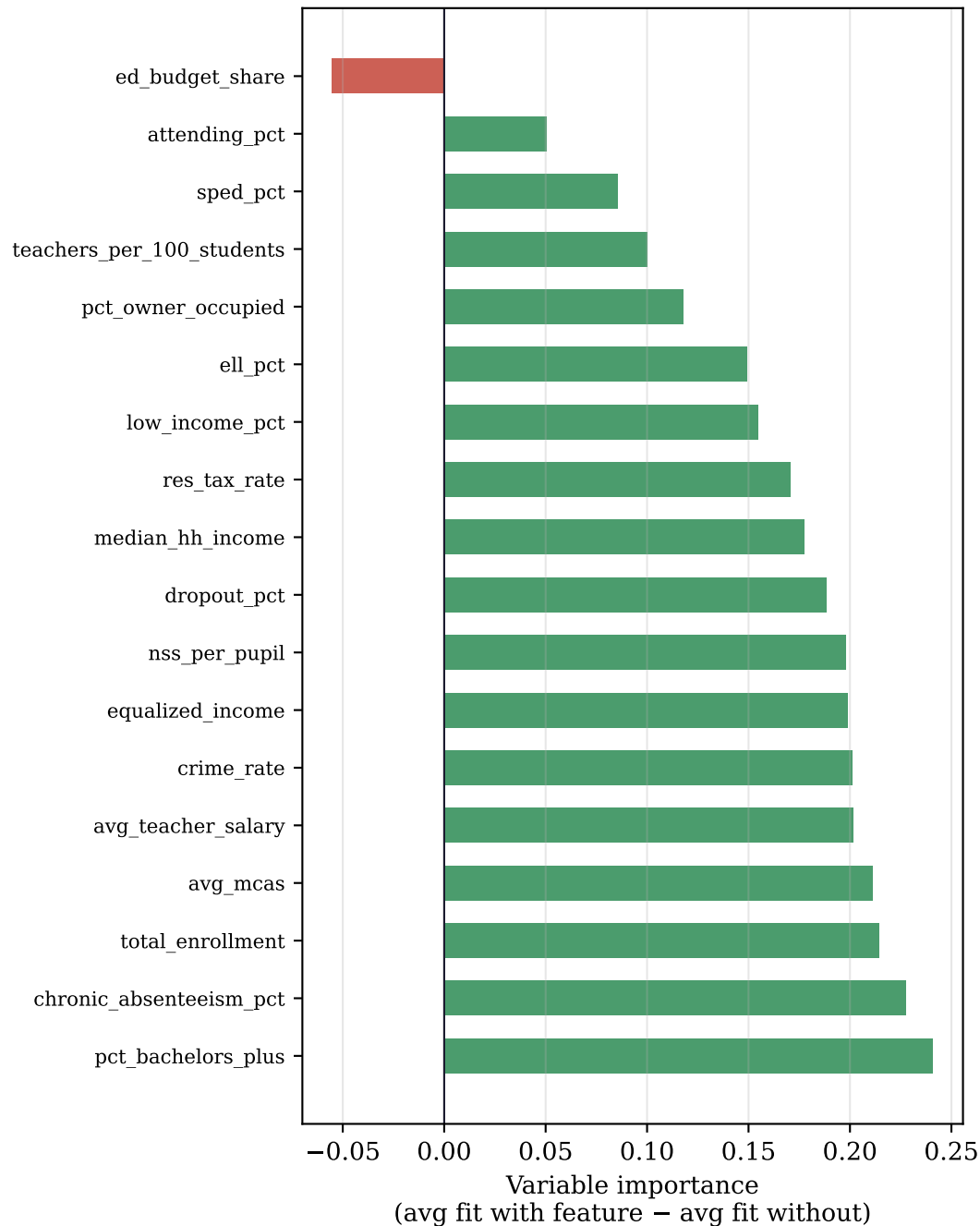
All Candidates — Variable Importance (all retained)
Green = helps (>0) · Grey ≈ 0 benign · Red < 0 harmful



Saugus RBP Analysis: MCAS Grade 10 (ELA)

Prediction task = Saugus · Predicted: 39.4pp Actual: 47.0pp Gap: +7.6pp (Fit: 0.1363)

Variable Importance (Exhibit 5)



Most Relevant Towns (highest weight)

Town	Weight	mcas10_ela
Everett	0.0748	0.37
Gardner	0.0674	0.40
Pittsfield	0.0622	0.40
Dracut	0.0615	0.53
Randolph	0.0532	0.36

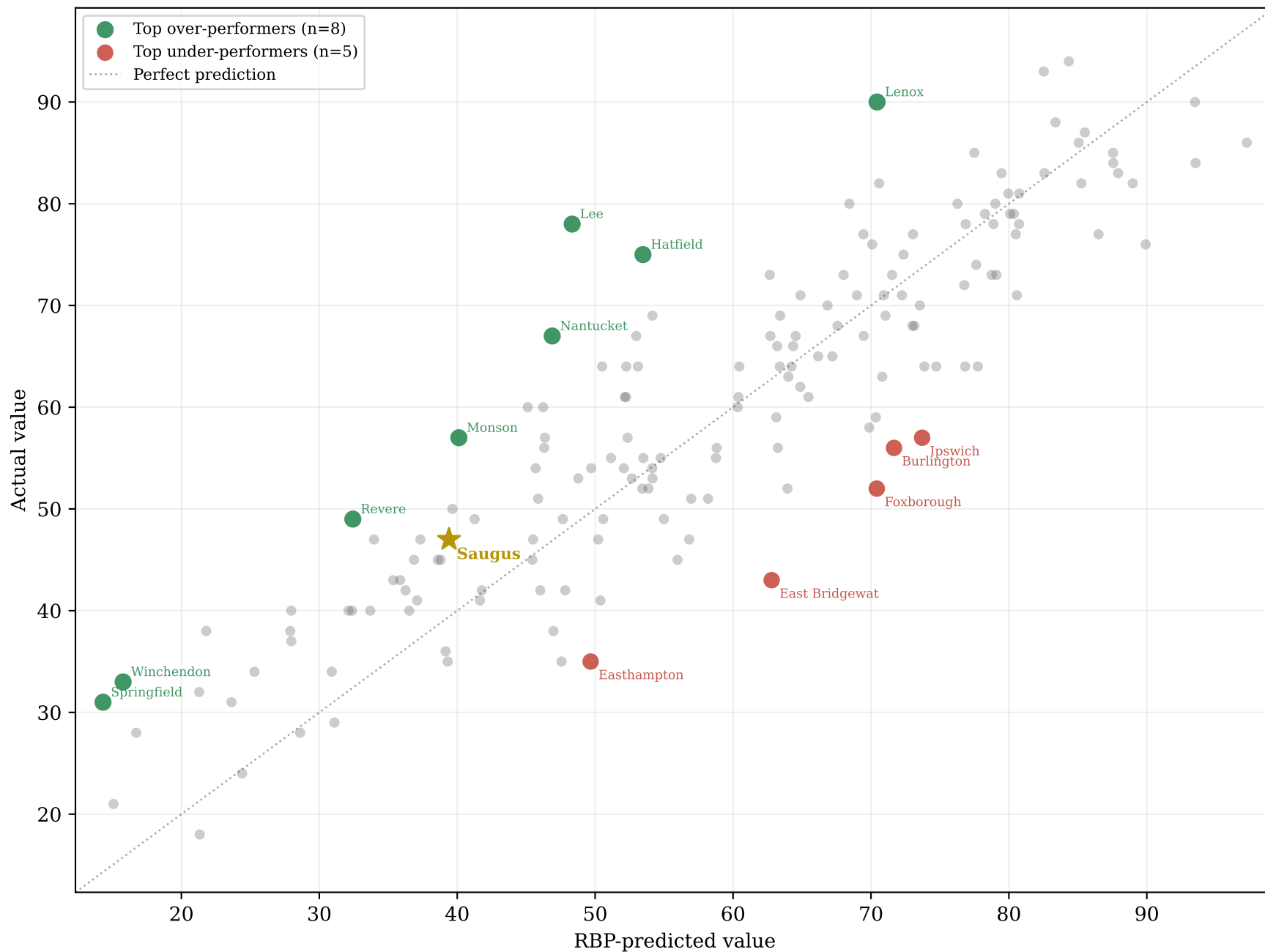
Least Relevant Towns (lowest weight)

Town	Weight	mcas10_ela
Andover	-0.0162	0.79
Foxborough	-0.0162	0.52
Hingham	-0.0157	0.86
Marshfield	-0.0156	0.76
Newburyport	-0.0153	0.71

Over- and Under-Performers: MCAS Grade 10 (ELA)

Saugus gap: +7.6pp · Gold star = Saugus · Green = over-performers · Red = under-performers

Actual vs Predicted — MCAS Grade 10 (ELA) (LOO r = 0.896)



Residual = actual - predicted. Positive = performing better than demographics suggest.

What Overachievers Do Differently: MCAS Grade 10 (ELA)

Feature values for top overachievers vs Saugus — ordered by RBP variable importance

Feature values — Saugus (highlighted) vs overachievers
Ordered by |importance|, top 15 shown

Feature	Saugus	Lee	Hatfield	Nantucket	Imp
pct_bachelors_plus	31.4	45.1	48.7	58.4	+0.241
chronic_absenteeism_pct	31.2	13.3	13.3	25.5	+0.228
total_enrollment	3,084	746.0	412.0	1,693	+0.214
avg_mcas	0.29	0.34	0.37	0.23	+0.211
avg_teacher_salary	89,778	82,282	87,392	102,575	+0.202
crime_rate	2,813	3,348	1,442	3,827	+0.201
equalized_income	1.17B	210M	132M	1.02B	+0.199
nss_per_pupil	17,369	24,155	24,753	28,994	+0.198
dropout_pct	2.90	1.00	0.00	1.50	+0.189
median_hh_income	100,819	74,222	100,381	119,750	+0.177
res_tax_rate	10.7	11.2	15.3	3.13	+0.171
low_income_pct	46.6	41.2	20.5	34.6	+0.155
ell_pct	13.0	5.60	0.30	19.0	+0.149
pct_owner_occupied	73.9	78.4	74.8	76.5	+0.118
teachers_per_100_student	5.87	9.69	9.59	9.43	+0.100

Overachiever residuals vs Saugus

District	Actual	Predicted	Gap
>> Saugus	47.0	39.4	+7.6pp
Lee	78.0	48.3	+29.7pp
Hatfield	75.0	53.5	+21.5pp
Nantucket	67.0	46.9	+20.1pp
Lenox	90.0	70.4	+19.6pp
Winchendon	33.0	15.8	+17.2pp
Monson	57.0	40.1	+16.9pp

Gap = actual – predicted. Positive = outperforming demographic expectation. Feature values shown in raw units; importance scores from RBP Exhibit 5.

What This Means: MCAS Grade 10 (ELA)

Predicted 39.4pp · Actual 47.0pp · Gap +7.6pp (higher is better) · Saugus ranks 41 of 168 MA districts on this measure

The Bottom Line

Saugus's grade 10 ELA proficiency is 47.0%, 7.6pp better than its demographic prediction of 39.4% — ranking 41 of 168 MA districts, in the top 25%. Grade 10 ELA reflects the cumulative academic base, not only current demographics.

What Shapes Saugus's Predicted Peer Group

Top RBP Exhibit 5 drivers of the Saugus prediction, vs. the statewide median:

Factor	Saugus	MA median	Importance
Adults with a bachelor's degree+	31.4%	46.9%	+0.24
Students chronically absent (10%+)	31.2%	15.8%	+0.23
Total district enrollment	3,084	1,669	+0.21
MCAS 3-8 (% meeting/exceeding)	29.0%	40.0%	+0.21
Average teacher salary	\$89,778	\$90,677	+0.20

Importance measures how much each factor sharpens the search for Saugus's true comparison districts — not whether higher values are good or bad for the outcome itself.

What Over-Performing Towns Do Differently

Same factors, for the 8 MA districts that most exceed their own prediction:

Factor	Saugus	Peer median	Difference
Adults with a bachelor's degree+	31.4%	37.1%	+5.6pp
Students chronically absent (10%+)	31.2%	21.9%	-9.3pp
Total district enrollment	3,084	1,135	-1,949
MCAS 3-8 (% meeting/exceeding)	29.0%	26.5%	-2.5pp
Average teacher salary	\$89,778	\$84,680	-5,098

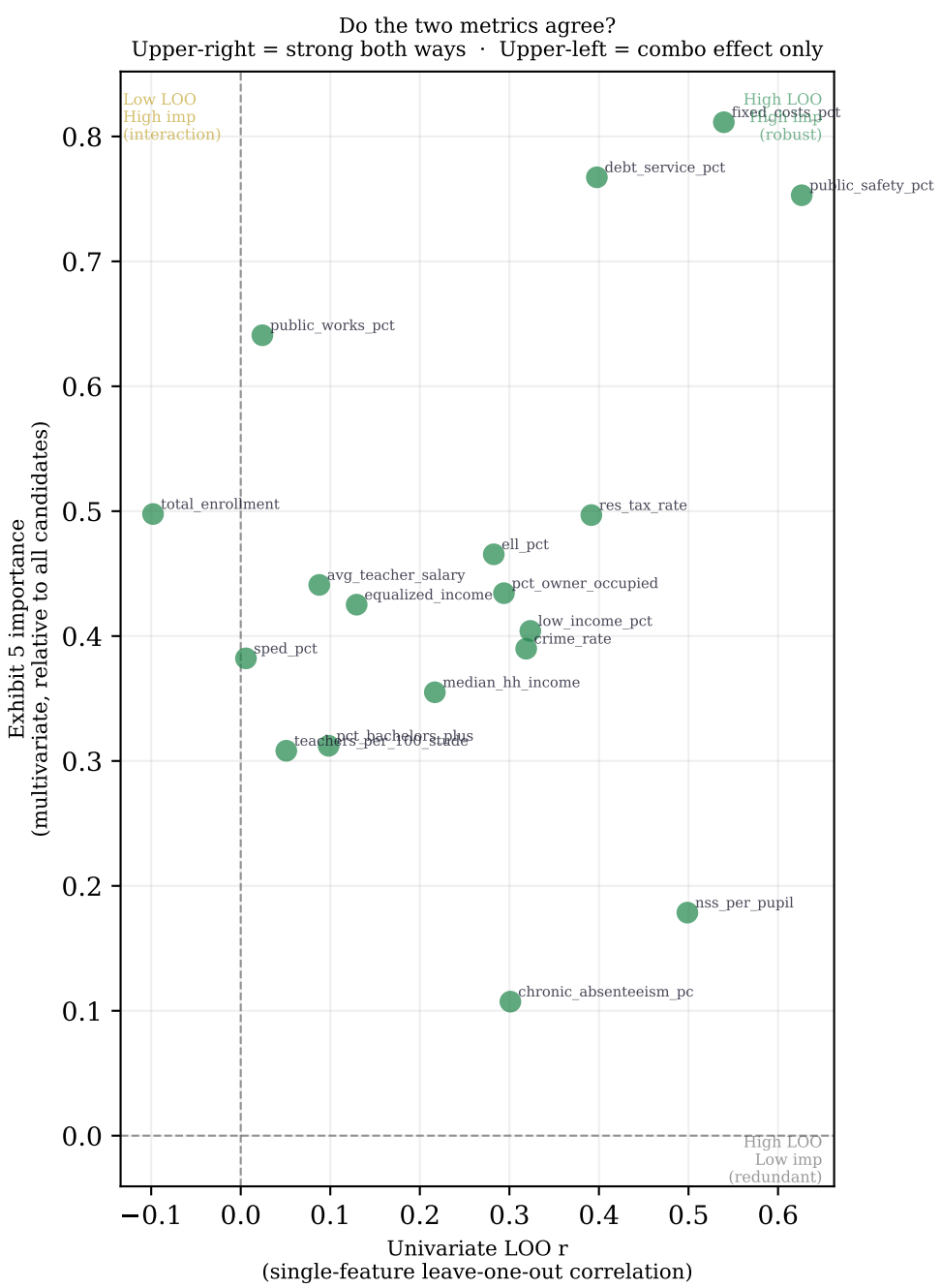
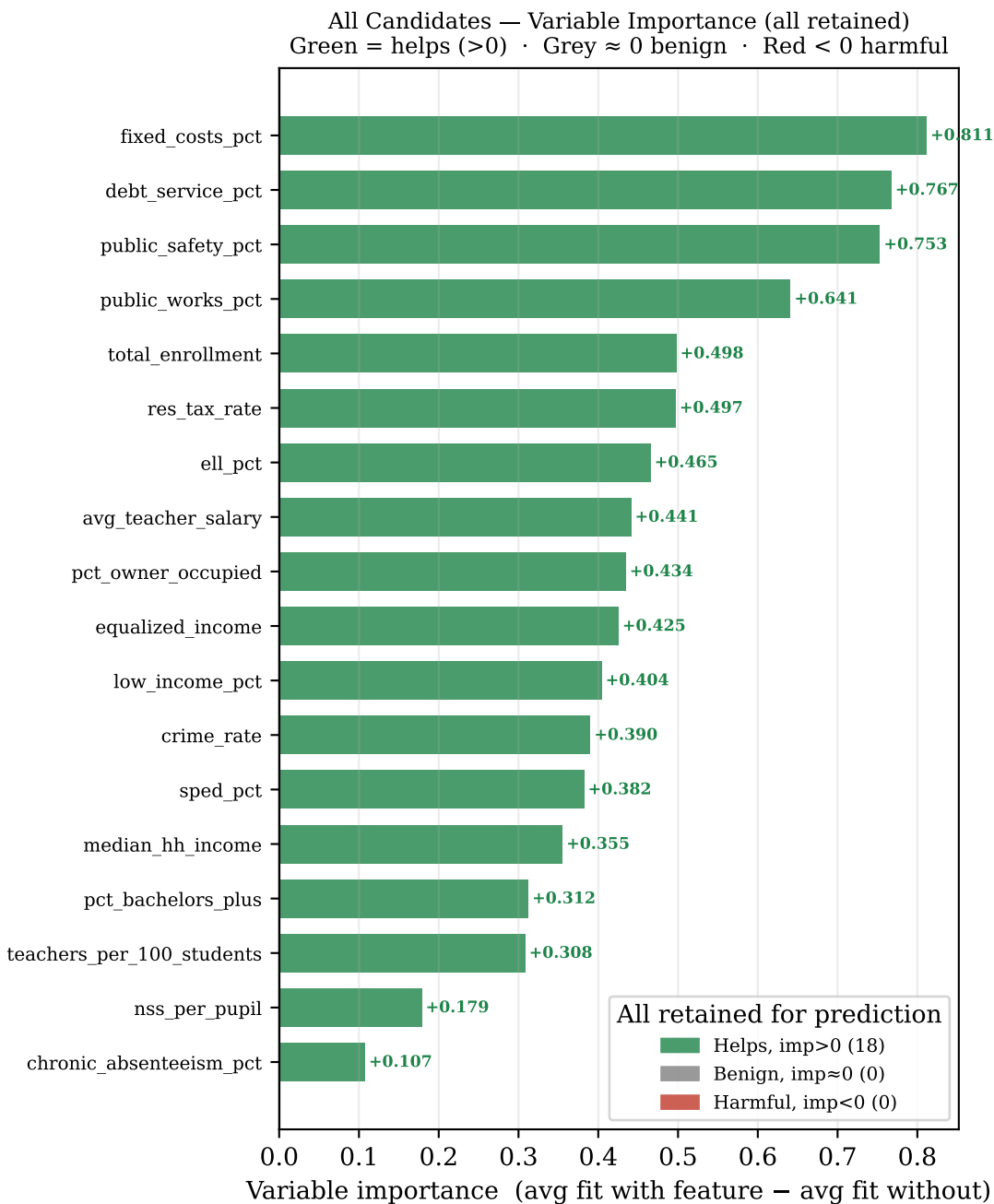
Difference = over-performer median – Saugus. Sign direction depends on the factor; see takeaway below.

The Takeaway

Saugus does better here than its demographic profile predicts. The strongest RBP driver of Saugus's predicted peer group is adults with a bachelor's degree+ (31.4% vs the 46.9% state median); the widest gap to the over-performing districts is average teacher salary (-5,098, peer median – Saugus). Saugus's grades 3-8 base, teacher compensation, and enrollment scale are the levers most visible among the drivers.

Variable Importance (Exhibit 5): Education Budget Share

18 candidates, all used for prediction · n_random=3000 · top-3 seed-stable · 18 help (imp>0) · 0 benign (≈0) · 0 harmful (imp<0)

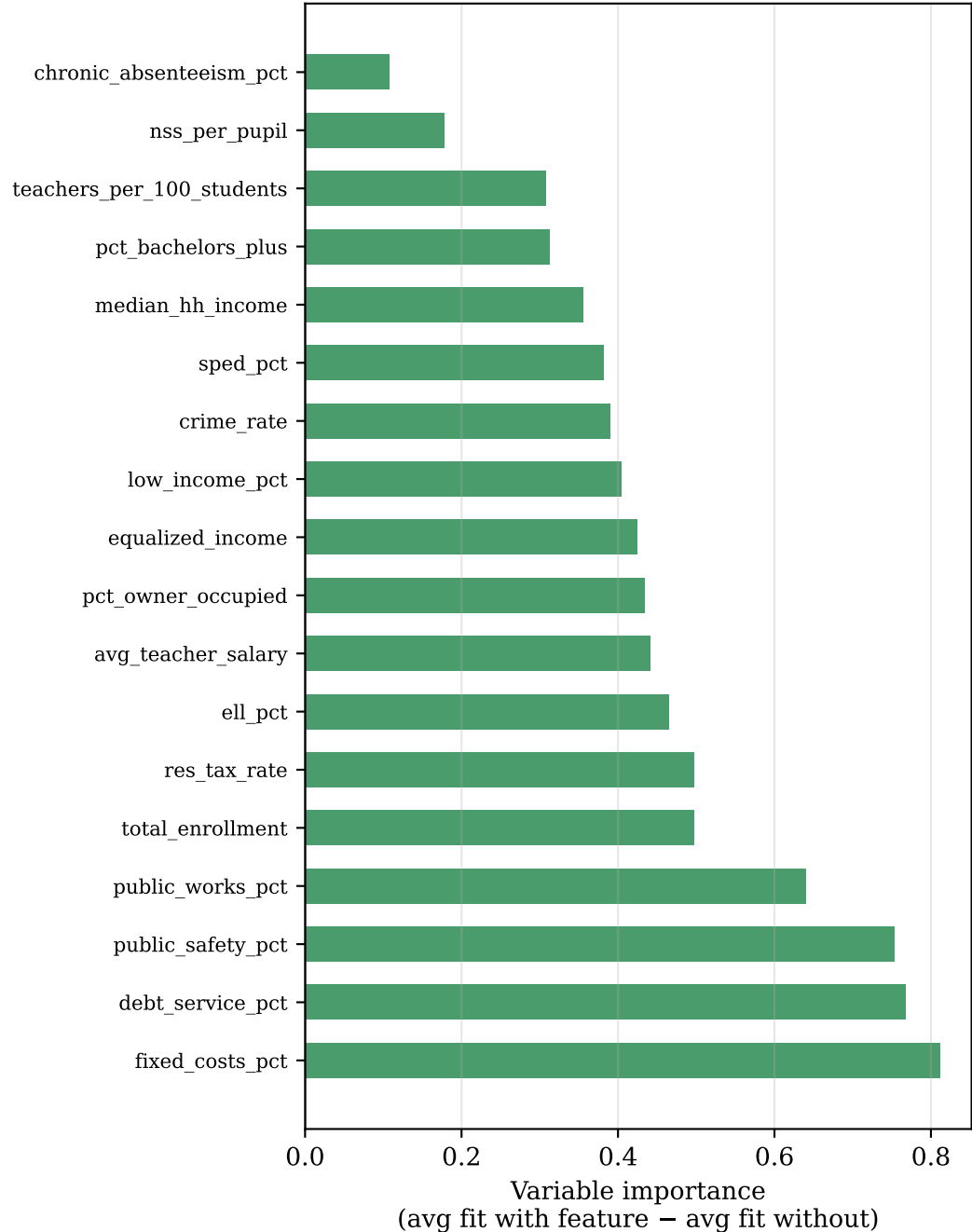


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Saugus RBP Analysis: Education Budget Share

Prediction task = Saugus · Predicted: 36.6pp Actual: 32.3pp Gap: -4.2pp (Fit: 0.1960)

Variable Importance (Exhibit 5)



Most Relevant Towns (highest weight)

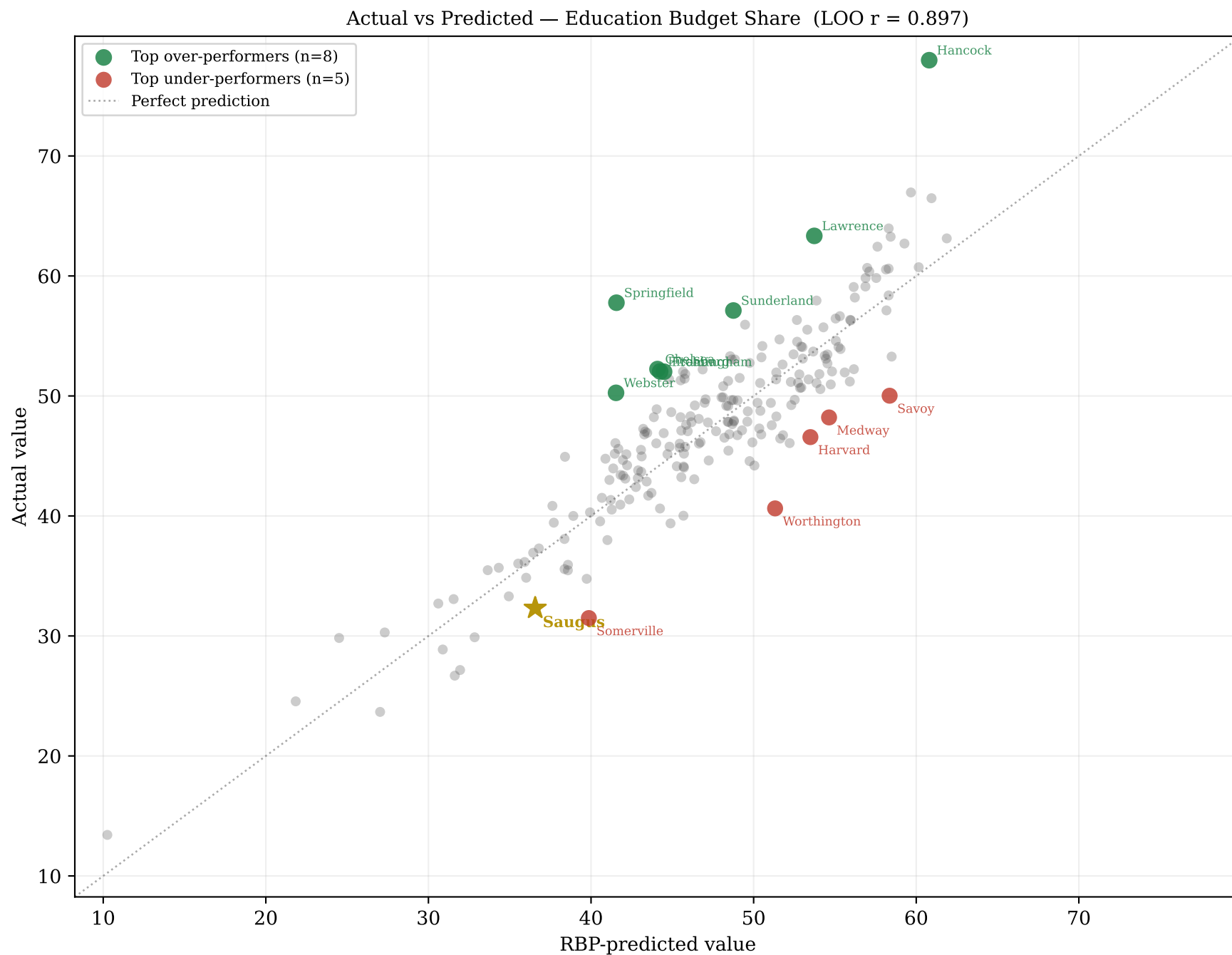
Town	Weight	ed_budget_sh
Gloucester	0.0519	33.30
Pittsfield	0.0488	40.53
Petersham	0.0438	56.65
Gardner	0.0417	43.68
Everett	0.0416	41.38

Least Relevant Towns (lowest weight)

Town	Weight	ed_budget_sh
Sunderland	-0.0173	57.12
Shutesbury	-0.0164	60.61
Amherst	-0.0152	52.21
Boxford	-0.0130	66.48
Ipswich	-0.0126	53.04

Over- and Under-Performers: Education Budget Share

Saugus gap: -4.2pp · Gold star = Saugus · Green = over-performers · Red = under-performers



Residual = actual - predicted. Positive = performing better than demographics suggest.

What Overachievers Do Differently: Education Budget Share

Feature values for top overachievers vs Saugus — ordered by RBP variable importance

Feature values — Saugus (highlighted) vs overachievers
Ordered by |importance|, top 15 shown

Feature	Saugus	Hancock	Springfie	Lawrence	Imp
fixed_costs_pct	25.3	2.43	19.7	7.45	+0.811
debt_service_pct	8.63	0.00	3.32	3.65	+0.767
public_safety_pct	15.2	4.61	11.0	8.66	+0.753
public_works_pct	7.35	6.46	1.35	3.98	+0.641
total_enrollment	3,084	106.0	29,468	15,500	+0.498
res_tax_rate	10.7	2.79	16.1	9.25	+0.497
ell_pct	13.0	0.00	17.1	42.1	+0.465
avg_teacher_salary	89,778	59,463	84,692	84,516	+0.441
pct_owner_occupied	73.9	77.8	49.5	30.3	+0.434
equalized_income	1.17B	13M	2.89B	1.77B	+0.425
low_income_pct	46.6	36.2	85.0	85.1	+0.404
crime_rate	2,813	0.00	8,923	3,435	+0.390
sped_pct	18.8	24.1	26.8	19.4	+0.382
median_hh_income	100,819	93,125	51,339	57,903	+0.355
pct_bachelors_plus	31.4	37.8	19.6	16.4	+0.312

Overachiever residuals vs Saugus

District	Actual	Predicted	Gap
>> Saugus	32.3	36.6	-4.2pp
Hancock	78.0	60.8	+17.2pp
Springfield	57.8	41.6	+16.2pp
Lawrence	63.3	53.7	+9.6pp
Webster	50.3	41.5	+8.7pp
Sunderland	57.1	48.8	+8.4pp
Chelsea	52.2	44.1	+8.2pp

Gap = actual – predicted. Positive = outperforming demographic expectation. Feature values shown in raw units; importance scores from RBP Exhibit 5.

What This Means: Education Budget Share

Predicted 36.6pp · Actual 32.3pp · Gap -4.2pp (higher is better) · Saugus ranks 202 of 219 MA districts on this measure

The Bottom Line

Saugus's education budget share is 32.3%, 4.2pp worse than its demographic prediction of 36.6% — ranking 202 of 219 MA districts, in the bottom 10%. Unlike the academic measures, this gap reflects a municipal budgeting choice, not a demographic constraint.

What Shapes Saugus's Predicted Peer Group

Top RBP Exhibit 5 drivers of the Saugus prediction, vs. the statewide median:

Factor	Saugus	MA median	Importance
Pension & benefits share of budget	25.3%	17.5%	+0.81
Debt service share of town budget	8.6%	5.0%	+0.77
Police & fire share of town budget	15.2%	11.8%	+0.75
Public works share of town budget	7.3%	5.3%	+0.64
Total district enrollment	3,084	1,669	+0.50

Importance measures how much each factor sharpens the search for Saugus's true comparison districts — not whether higher values are good or bad for the outcome itself.

What Over-Performing Towns Do Differently

Same factors, for the 8 MA districts that most exceed their own prediction:

Factor	Saugus	Peer median	Difference
Pension & benefits share of budget	25.3%	17.6%	-7.7pp
Debt service share of town budget	8.6%	3.0%	-5.6pp
Police & fire share of town budget	15.2%	10.7%	-4.5pp
Public works share of town budget	7.3%	4.0%	-3.3pp
Total district enrollment	3,084	6,594	+3,510

Difference = over-performer median – Saugus. Sign direction depends on the factor; see takeaway below.

The Takeaway

Saugus does worse here than its demographic profile predicts. The strongest RBP driver of Saugus's predicted peer group is pension & benefits share of budget (25.3% vs the 17.5% state median); the widest gap to the over-performing districts is total district enrollment (+3,510, peer median – Saugus). The top drivers are competing budget categories that crowd out the education share — the lever here sits with Town Meeting and municipal budgeting, not the school district.

Combined Model Summary: All Four Outcomes

RBP factor selection results across MCAS grades 3–8, Dropout Rate, MCAS grade 10 ELA, and Education Budget Share

Model	Predicted	Actual	Gap	Direction	LOO r	Features
MCAS Grades 3–8	21.8%	29.0%	+7.2pp	over	+0.944	17
Dropout Rate	3.7%	2.9%	-0.8pp	over	+0.831	18
MCAS Grade 10 (ELA)	39.4%	47.0%	+7.6pp	over	+0.896	18
Education Budget Share	36.6%	32.3%	-4.2pp	under	+0.897	18

Feature Cross-Reference — Which Factors Earn Their Place Across Models

A feature appearing in multiple models is a robust signal, not a model-specific artifact.

Feature	# Models	MCAS 3-8	Dropout	MCAS 10	Ed Budget
chronic absenteeism pct	4	+0.367	+0.558	+0.228	+0.107
total enrollment	4	+0.315	+0.188	+0.214	+0.498
equalized income	4	+0.288	+0.210	+0.199	+0.425
avg teacher salary	4	+0.247	+0.227	+0.202	+0.441
pct bachelors plus	4	+0.353	+0.204	+0.241	+0.312
crime rate	4	+0.253	+0.239	+0.201	+0.390
res tax rate	4	+0.256	+0.139	+0.171	+0.497
ell pct	4	+0.264	+0.166	+0.149	+0.465
pct owner occupied	4	+0.302	+0.136	+0.118	+0.434
low income pct	4	+0.255	+0.177	+0.155	+0.404
median hh income	4	+0.194	+0.238	+0.177	+0.355
sped pct	4	+0.227	+0.168	+0.086	+0.382
teachers per 100 students	4	+0.156	+0.177	+0.100	+0.308
nss per pupil	4	+0.179	+0.086	+0.198	+0.179
ed budget share	3	+0.062	-0.333	-0.055	—
attending pct	3	+0.189	+0.037	+0.050	—
dropout pct	2	+0.298	—	+0.189	—
avg mcas	2	—	+0.218	+0.211	—
fixed costs pct	1	—	—	—	+0.811
debt service pct	1	—	—	—	+0.767
public safety pct	1	—	—	—	+0.753
public works pct	1	—	—	—	+0.641
mcas10_ela	1	—	+0.210	—	—

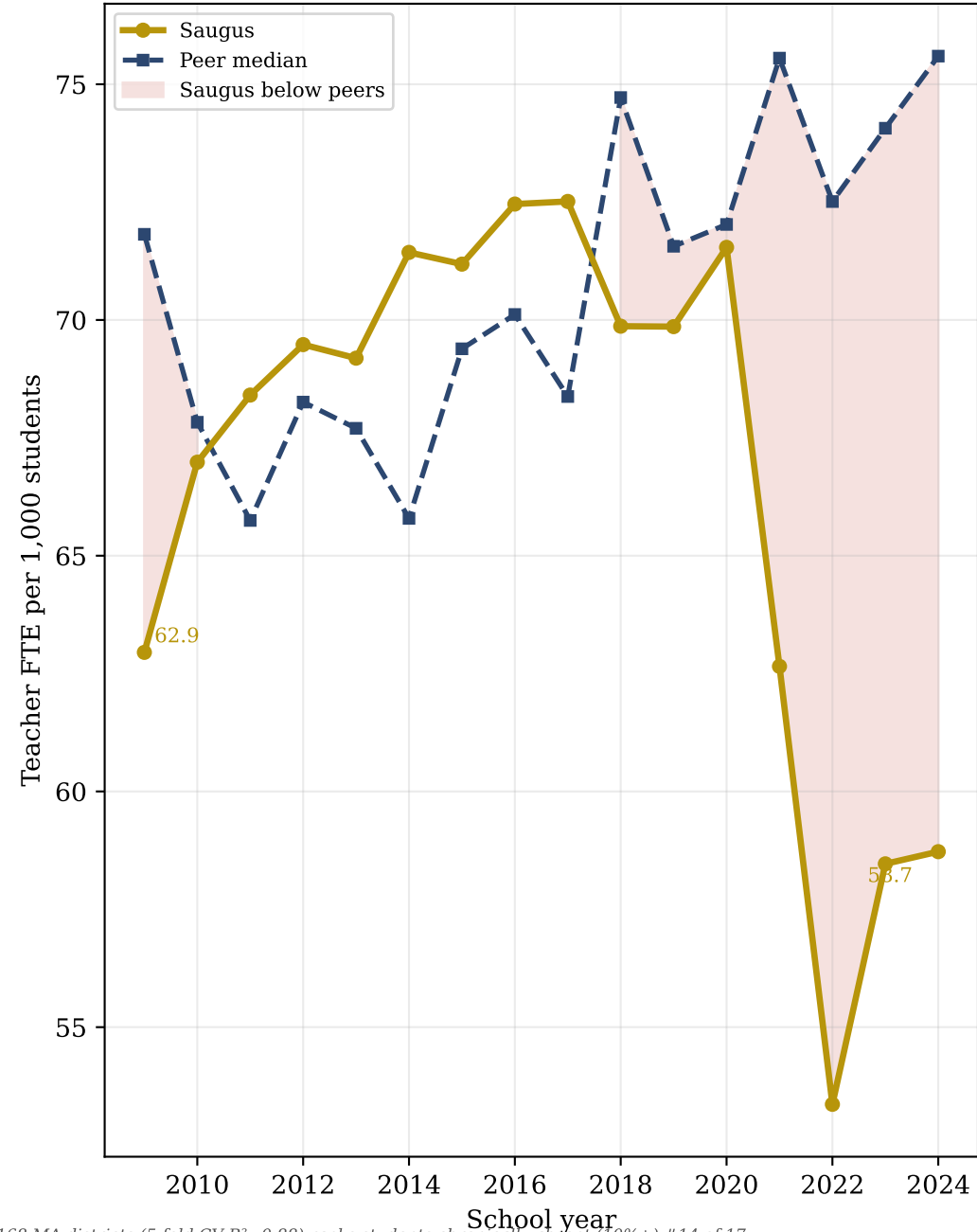
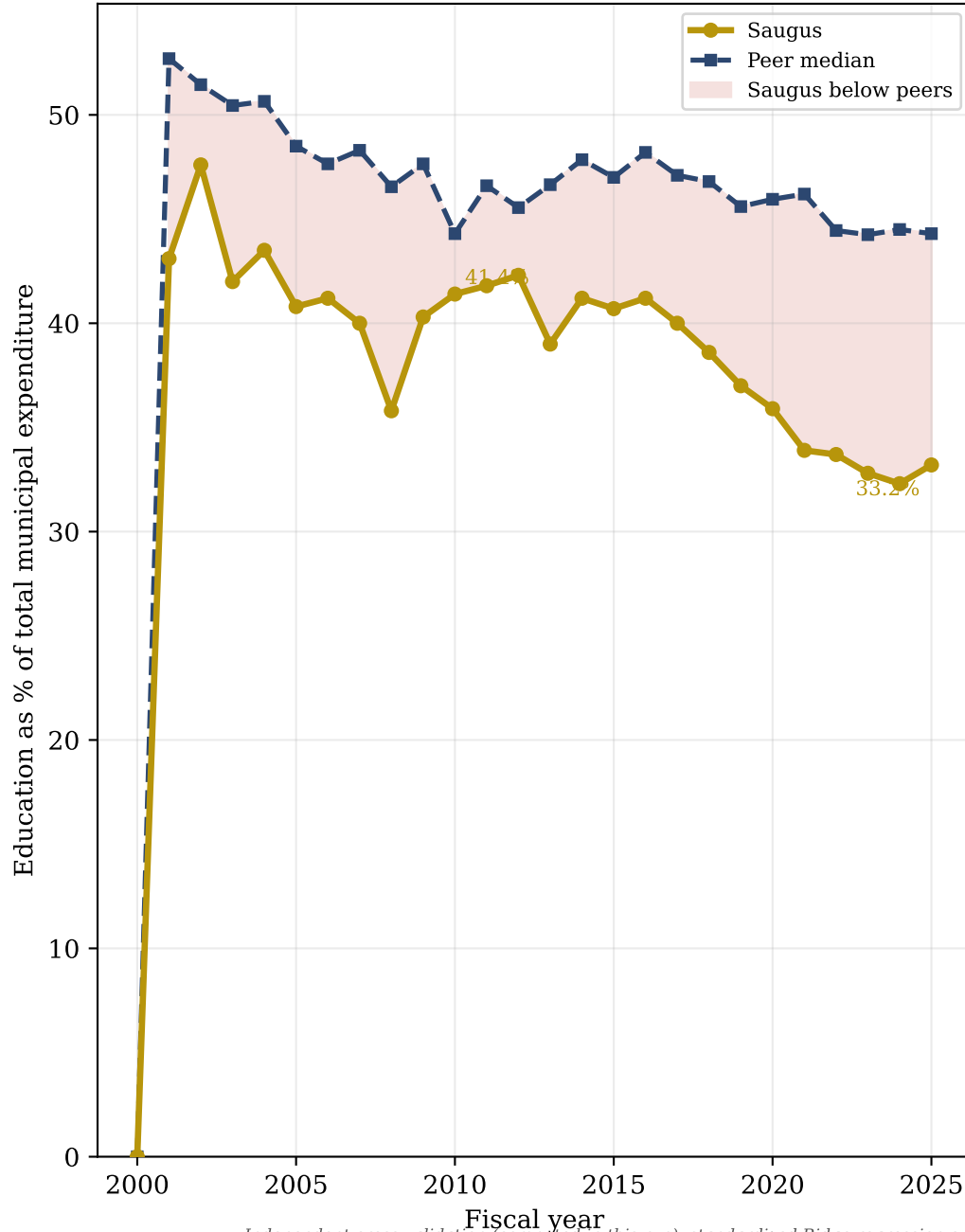
Green rows = feature appears in 3+ models. Cell colour: green = positive importance, red = negative importance, white = not in that model (—).

Budget Allocation & Teacher Density — 15-Year Trajectory

Source: MA DLS Schedule A (budget) · MA DESE Staffing (teachers) · Peer comparison: 8 demographically similar overachiever towns

Education Budget Share
(Schedule A, general fund)

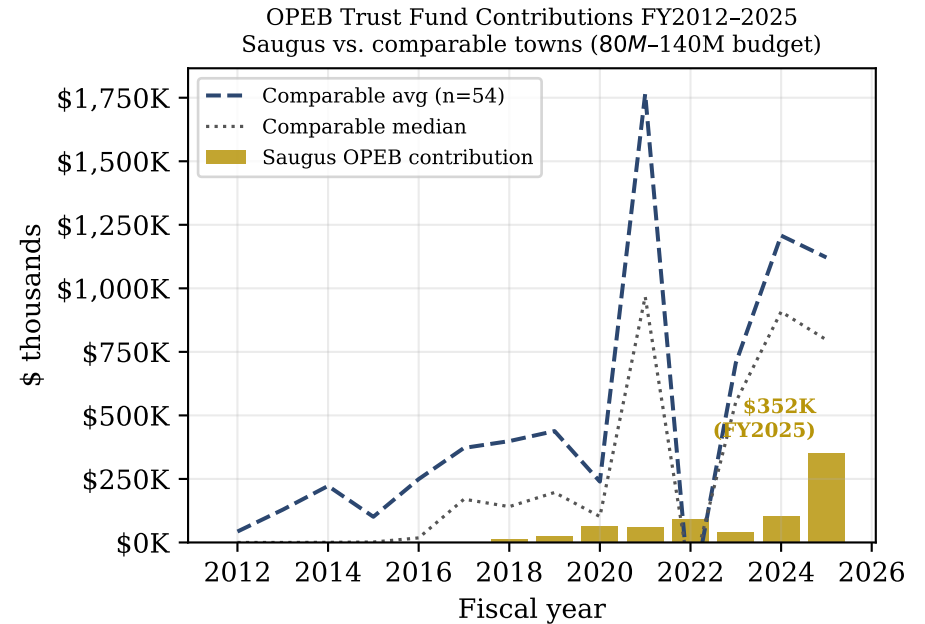
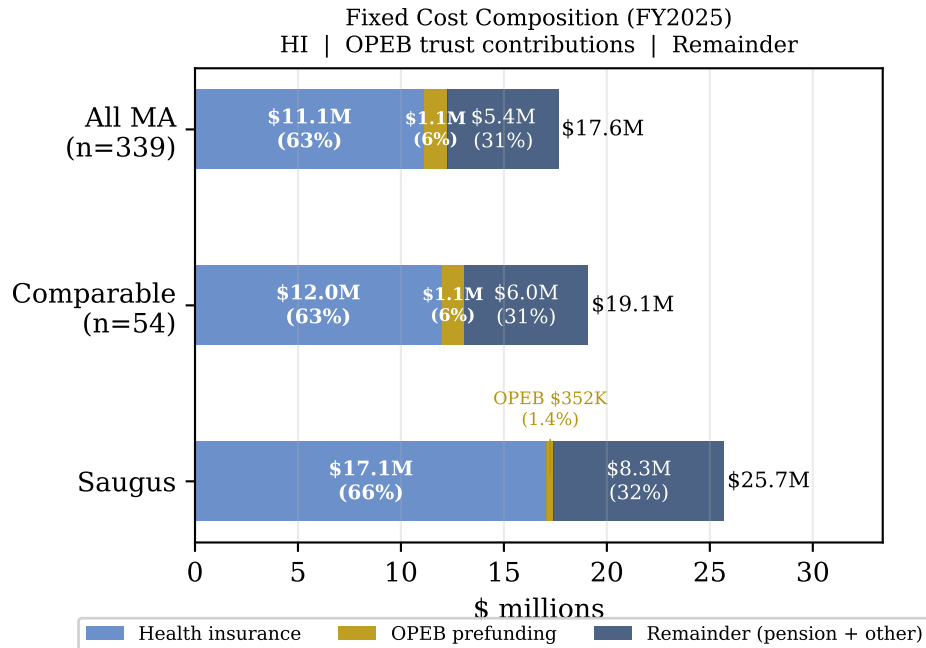
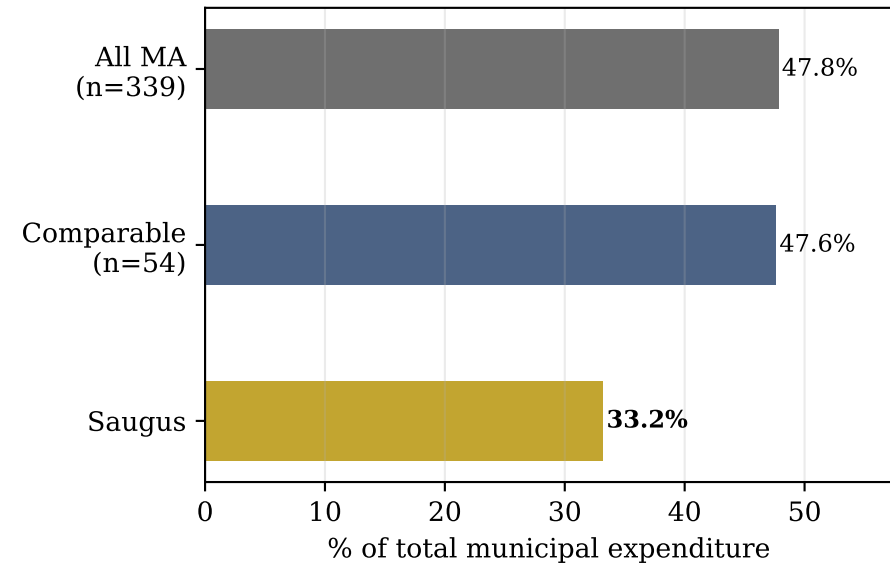
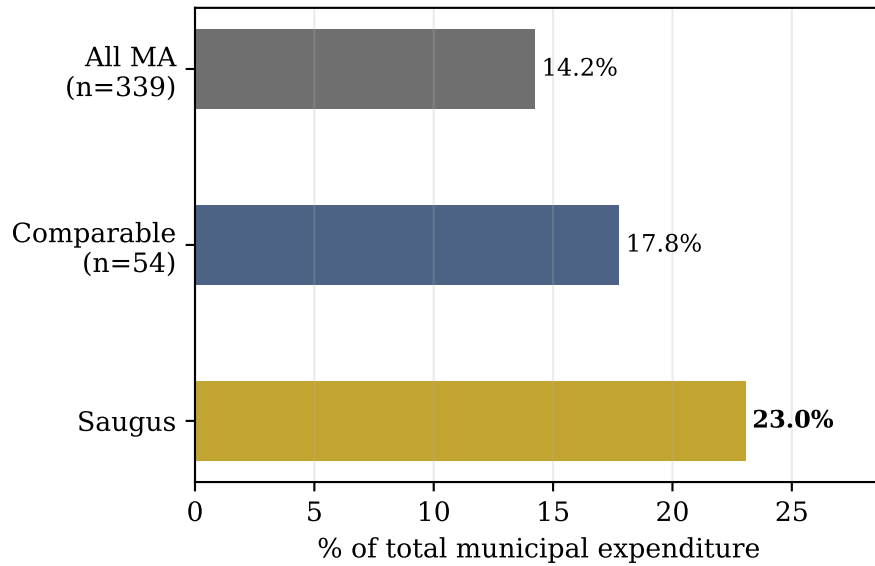
Teacher Density
(DESE Staffing + Enrollment)



Independent cross-validation (computed in this run): standardized Ridge regression on 168 MA districts (5-fold CV $R^2=0.88$) ranks students chronically absent (10%+) #14 of 17 standardized predictors of MCAS 3–8 ($|standardized \beta|=0.04$) — a divergence, not corroboration: a linear model assigns students chronically absent (10%+) a small standardized coefficient, whereas RBP's Exhibit 5 ranks it among the top drivers — the kind of nonlinear / interaction effect RBP is designed to surface and linear regression misses. Peer set: 8 demographically similar overachiever towns (Mahalanobis $d_M \leq 2$).

Fixed Costs: Sources, Scale, and OPEB Prefunding

FY2025 · Comparable = 54 MA towns (80M-140M) · All MA = 339 · Columns sum exactly to total for all 4,898 statewide municipality-years



Optimum Profile — What Overachievers Look Like

Each grid shows actionable feature values for Saugus vs overachiever towns. Showing both peer groups tests whether the finding holds across definitions.

Overachievers — all 20 MA districts that beat their demographic prediction

Feature	Saugus	Springfie	Quincy	Winchendo	Lawrence	Chelsea	Peer med	Gap	N
Chronic absenteeism (%)	31.2%	32.4%	16.5%	35.6%	29.2%	35.4%	23.8%	-7.4pp	20
Teachers / 100 students	5.9	7.4	7.5	6.7	7.4	6.7	7.5	+1.6	20
Avg teacher salary	\$89,778	\$84,692	\$94,955	\$83,722	\$84,516	\$89,573	\$87,302	\$-2,476	20
Net school spending / pupil	\$17,369	\$26,616	\$20,257	\$19,559	\$23,178	\$22,535	\$22,390	+\$+5,020	20
Shown: Springfield, Quincy, Winchendon, Lawrence, Chelsea (+ 15 more in median)									

Demographically Similar Overachievers — 16 towns within 2σ Mahalanobis distance of Saugus on income, property wealth, poverty, size, education & ELL

Feature	Saugus	Norwood	Clinton	Fairhaven	West Boyl	Sandwich	Peer med	Gap	N
Chronic absenteeism (%)	31.2%	25.1%	7.7%	15.6%	15.6%	17.9%	20.9%	-10.2pp	16
Teachers / 100 students	5.9	8.8	7.3	7.5	8.1	8.1	7.8	+2.0	16
Avg teacher salary	\$89,778	\$92,324	\$87,059	\$82,224	\$89,252	\$87,840	\$87,302	\$-2,476	16
Net school spending / pupil	\$17,369	\$21,084	\$18,512	\$17,287	\$19,718	\$26,445	\$20,975	+\$+3,606	16
Shown: Norwood, Clinton, Fairhaven, West Boylsto, Sandwich (+ 11 more in median)									

Similarity metric: Mahalanobis distance $d_M = \sqrt{(\Delta x \cdot \Sigma^{-1} \cdot \Delta x')}$, Σ = sample covariance of all MA districts on (median_hh_income, equalized_income, low_income_pct, total_enrollment, pct_bachelors_plus, ell_pct). Threshold: $d_M \leq 2$ (2σ). Pre-specified; not tuned to result. Gap = peer median – Saugus. Red = Saugus below peer median, Green = at or above.